

ALAN S

Electronics & Communication Engineering Student

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PROFESSIONAL SUMMARY

Electronics and Communication Engineering student with hands-on experience in embedded systems, IoT, sensor interfacing, electronics, PCB design, robotics, telecommunications, and scientific data processing. Experienced in developing ESP32-based systems, working with communication technologies, processing scientific datasets, and applying Python and machine learning to engineering problems. Interested in embedded systems, electronics, telecommunications, space technology, and aerospace applications.

EDUCATION

Government Engineering College, Idukki | B.Tech — Electronics & Communication Engineering

Expected Graduation: 2027

INTERNSHIP EXPERIENCE

Asianet Satellite Communications Ltd. — Internship Trainee, Network Management Department

June 2026 – July 2026

- Completed industrial internship training in the Network Management Department.
- Gained exposure to telecommunications and network-management operations.
- Developed practical understanding of industry-level communication infrastructure.

STEM Robotics International / SRAI Smart Solutions Pvt. Ltd. — AI & Robotics Intern

July 2024 | 7-day internship

- Completed a 7-day internship focused on AI and robotics.
- Participated in robotics-related projects and technical activities.
- Gained practical exposure to AI and robotics applications.

PROJECTS

Intelligent IoT-Based Multi-Hazard Disaster Early Warning System

Technologies: ESP32, Sensors, IoT, Wi-Fi, LoRa, Python, Flask, Machine Learning

- Developed an IoT-based multi-hazard monitoring system using distributed ESP32 sensor nodes.
- Integrated sensors for soil moisture, vibration, temperature, humidity, flame/gas and environmental monitoring.
- Developed a data-collection and processing pipeline using Python and Flask.
- Implemented machine-learning-based risk classification for disaster early-warning applications.
- Designed a web-based monitoring and visualization system for sensor data and risk conditions.

Embedded / LoRa Communication Project

Technologies: ESP32, Arduino, LoRa RA-01H, Embedded C, Sensors

- Developed wireless communication between embedded nodes using LoRa.
- Interfaced sensors with microcontrollers for real-time data acquisition.
- Developed embedded C firmware for sensor measurement and communication.
- Tested hardware communication and embedded-system integration.

TECHNICAL SKILLS

Programming: C, Embedded C, Python

Embedded Systems: ESP32, Arduino, Microcontrollers, Sensor Interfacing, GPIO, ADC, PWM, UART, I²C, SPI

Electronics & Hardware: Digital Electronics, Analog Electronics, Circuit Debugging, Hardware Testing, PCB Design

Communication & IoT: Wi-Fi, LoRa, IoT, Wireless Communication, Network Management Fundamentals

PCB & Development Tools: KiCad, Arduino IDE, PlatformIO, VS Code

FPGA / Digital Design: Verilog, RTL Design, FPGA Fundamentals

Data & Machine Learning: NumPy, Pandas, Matplotlib, Scikit-learn

Development Environment: Git / GitHub, Jupyter Notebook

WORKSHOPS & TRAINING

DroneX — One-Day Hands-on Drone Workshop | IEEE MTTs SBC GECI & IEEE RAS SBC GECI | February 2026

Practical exposure to drone and UAV-related technology through a hands-on technical workshop.

SOFT SKILLS

Problem Solving • Technical Communication • Analytical Thinking • Debugging & Troubleshooting • Team Collaboration • Self-Learning

CAREER FOCUS

Embedded Systems | Electronics & PCB Design | IoT & Hardware Development | Telecommunications | Robotics | Space & Aerospace Electronics | Digital Electronics | FPGA / RTL Design